Scarlett H. Francini

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EDUCATION

Duke University Pratt School of Engineering, Durham, NC | GPA: 3.76/4.0 Expected May 2027
BSE in Electrical and Computer Engineering and Computer Science

- Relevant Coursework: Digital Systems, Computer Architecture, Data Structures, Probability, French Practicum
- Benjamin N. Duke (BN) Full-ride Merit Scholarship Recipient and 2023 GSUSA Gold Award Scholarship Recipient
- Associations: Duke Technology Scholar, Merit Scholars Board, SWE, IEEE, ECE 280 Signals and Systems TA

North Carolina School of Science and Mathematics, Durham, NC | GPA: 5.0/4.0 | SAT: 1560 August 2021 – May 2023

- Selected for Duke Health & Environment (2022) and Harvard Math & Social Justice (2023) summer programs

SKILLS

- **Languages:** Python, Java, C++, C, JavaScript, Verilog, MATLAB, R, HTML/CSS
- **Systems & Technologies:** React/Vite, FastAPI, Celery, Docker, Git, Linux, Jupyter, Pandas, NumPy, n8n, NVIDIA Sionna, PDAL, Geo2SigMap, NSF FABRIC, UDP networking, FPGA, ROS, MIPS

TECHNICAL EXPERIENCE

RENCI | Cyberinfrastructure Intern (NSF REU) May 2026 – Present

- Building a 2-node Fermilab-to-NERSC data streaming and visualization pipeline on the NSF FABRIC testbed, using Python/Jupyter workflows to replay DUNE-style packet data and generate visualization-ready outputs

Timao, Marseille, France | AI Consultant June – July 2025

- Engineered Python and n8n workflows integrating custom GPTs and retrieval-augmented generation (RAG) to parse, extract, summarize, and classify French procurement documents, reducing review time by 93%

Duke University FuNCtions Lab | Wireless Systems Research Assistant December 2024 – May 2026

- Co-developed ClickDT, a 5-module platform integrating LiDAR and OpenStreetMap data to generate high-resolution 3D wireless simulation scenes.

Duke University Nanomaterials and Thin Films Lab | Software & Data Research Assistant June 2022 – August 2024

- Built Python and MATLAB GUIs for instrument control and data analysis for an underwater mass spectrometer
- Co-authored 2 peer-reviewed publications: “Spectral Reconstruction Improvement in a Cycloidal Coded-Aperture Mass Spectrometer” in American Chemical Society (Apr 2024), and “A Super-Resolution Coded Aperture Miniature Mass Spectrometer Proof-of-Concept for Planetary Science” in International Journal of Mass Spectrometry (Jan 2025)

Harvest Hope Food Bank 501(c)(3), Columbia, SC | Duke BN Data Volunteer Intern May – June 2024

- Digitized 100+ grant records, improving data accessibility and accelerating food distribution across 20 SC counties

FIRST Robotics Competition, Team 900 | Programming Team Member August 2021 – May 2023

- Led URDF and C++ software development, contributing to 2023 World Championship Innovation in Control Award

PROJECTS

- **Credit Suisse:** Presented portfolio recommendation via Tableau for optimal returns with Python analysis
- **ClickDT:** Built full-stack features for scalable 3D digital twin generation, developing frontend UI/UX and backend (Celery, PDAL, Geo2SigMap) integration with LiDAR/OSM data; demo presented at IEEE MILCOM
- **Markov & Huffman:** Implemented Java algorithms for text generation and file compression/decompression
- **JukeBox:** Built FPGA jukebox with custom Verilog CPU, RFID/coin inputs, memory-mapped I/O, and PWM audio
- **Girl Scouts Gold Award:** Implemented a self-sustaining composting program at The Hawbridge School